

Ascham School



Mathematics Extension 1

Trial HSC Examination

Wednesday 29th July 2020

2 hours

General Instructions • Reading time – 10 minutes

- Working time – 2 hours
- Write using black non-erasable pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- In Questions 11–14, show relevant mathematical reasoning and/ or calculations

Total marks:

70

Section I – 10 marks (pages 2–3)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 4–7)

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section

Section I

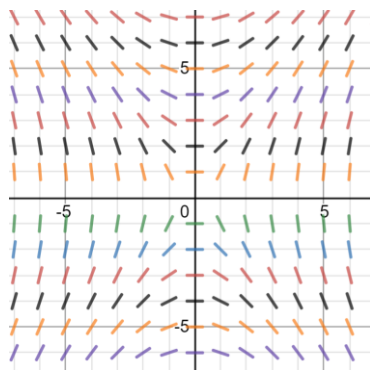
10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 The slope field below could represent which of the following differential equations?

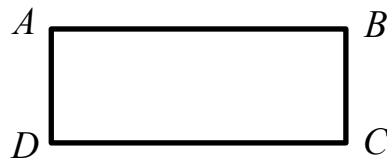


- A. $\frac{dy}{dx} = \frac{2x}{y}$ B. $\frac{dy}{dx} = \frac{2y}{x}$ C. $\frac{dy}{dx} = \frac{x^2}{y^2}$ D. $\frac{dy}{dx} = \frac{y^2}{x^2}$
- 2 Each of the letters A, A, C, H, M, and S is written on a separate card. The cards are drawn at random from a hat and placed next to each other to form a word. What is the probability that the word ASCHAM appears?
- A. $\frac{1}{60}$ B. $\frac{1}{360}$ C. $\frac{1}{720}$ D. $\frac{1}{50}$
- 3 The solution for the inequality $\frac{3}{x-1} < 5$ can be expressed as:
- A. $x \in \left[\frac{8}{5}, \infty\right)$ B. $x \in (-\infty, 1] \cup \left[\frac{8}{5}, \infty\right)$
 C. $x \in \left(\frac{8}{5}, \infty\right)$ D. $x \in (-\infty, 1) \cup \left(\frac{8}{5}, \infty\right)$
- 4 A possible solution for $\tan \frac{\theta}{2}$ given $2 \cos \theta + \sin \theta = -1$ is:
- A. -3 B. $-\frac{1}{3}$ C. $\frac{1}{3}$ D. 3

5 $\int \frac{x}{\sqrt{1-x^2}} dx =$

- A. $\sin^{-1} x + c$ B. $-x \cos^{-1} x + c$ C. $-\frac{1}{2}\sqrt{1-x^2} + c$ D. $-\sqrt{1-x^2} + c$

6 Consider the rectangle below.



Which of the following statements is **false**?

- A. $\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$ B. $\overrightarrow{AB} - \overrightarrow{BC} = \overrightarrow{DB}$
 C. $\overrightarrow{AB} - \overrightarrow{BC} = \overrightarrow{BD}$ D. $\overrightarrow{AB} \bullet \overrightarrow{BC} = 0$

7 A projectile has the equation of path $y = -3x^2 + 2x + 4$. How far will it have travelled horizontally before it returns to its original height?

- A. $\frac{1}{3}$ unit B. $\frac{2}{3}$ unit C. 2 units D. 4 units

8 The coefficient of the fourth term in the expansion of $(3x-4)^6$ is:

- A. 34560 B. -34560 C. 25920 D. -25920

9 The graph of the function $y = \cos^{-1}(2x)$ is dilated horizontally by a dilation factor of 4 and then translated vertically by 3 units. What is its new equation?

- A. $y = \cos^{-1}\left(\frac{2x}{4} + 3\right)$ B. $y = \cos^{-1}\left(\frac{2x}{4}\right) + 3$
 C. $y = \cos^{-1}(8x) + 3$ D. $y = \cos^{-1}(8x + 3)$

10 5 boys and 5 girls are seated at a round table. In how many ways can this happen with Robert and Hannah sitting together?

- A. 725760 B. 362880 C. 80640 D. 40320

Section II**60 marks****Attempt Questions 11–14****Allow about 1 hour and 45 minutes for this section**

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use the Question 11 Writing Booklet.

(a) Given $\underline{a} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$ and $\underline{b} = \begin{pmatrix} 4 \\ 7 \end{pmatrix}$

(i) Find $\underline{a} + \underline{b}$ and $\underline{a} - \underline{b}$ 2

(ii) Hence, find $(\underline{a} + \underline{b}) \cdot (\underline{a} - \underline{b})$. State if $\underline{a} + \underline{b}$ is perpendicular to $\underline{a} - \underline{b}$ and justify your answer with a reason. 2

(b) The polynomial $P(x) = x^3 - 4x^2 + tx + 10$ has roots $\alpha, -\alpha$ and β . Find the three roots and hence find the value of t . 3

(c) Zara is testing a batch of homemade bullets on her shooting range. She holds her rifle at shoulder height (170 cm above the ground) and shoots horizontally with an initial speed of 360 m/s at a target 100m away. Assume acceleration due to gravity is 10m/s^2 .

(i) Find expressions for $\dot{y}$ and $\dot{x}$. 2

(ii) Find expressions for y and x . 2

(iii) Hence find the equation of path as y in terms of x . 1

(iv) Assuming the ground is horizontal at her range, how far above the ground will the bullet hit the target? Give your answer correct to the nearest cm. 1

(v) When does the bullet hit her target? Find the impact speed (correct to 5 significant figures). 2

End of Question 11

Question 12 (15 marks) Use the Question 12 Writing Booklet.

- (a) Find the particular solutions to the differential equation $\frac{dy}{dx} = \frac{2x-1}{y^2}$ given **3**

$$y = \sqrt[3]{3} \text{ when } x = 0.$$

- (b) Solve $3 \cos 3x - 4 \sin 3x = 5$ for $0 \leq x \leq \frac{2\pi}{3}$. **3**

- (c) The area bounded by the line $y = x - 1$ and the curve $y = 4(x - 1)^2$ is rotated about the x -axis.

- (i) Sketch the bounded area, noting any intercepts. **2**

- (ii) Find the volume generated. **3**

- (d) Given $\underline{u} = 2\underline{i} + 3\underline{j}$ and $\underline{v} = -2\underline{i} + 4\underline{j}$,

- (i) Find $\text{proj}_{\underline{u}} \underline{v}$ **2**

- (ii) Find $\underline{w} = \underline{v} - \text{proj}_{\underline{u}} \underline{v}$ **1**

- (iii) What is the angle between $\underline{u}$ and $\underline{w}$? **1**

End of Question 12

Question 13 (15 marks) Use the Question 13 Writing Booklet.

- (a) Prove by mathematical induction: 3

$$a + ar + ar^2 + ar^3 + \dots + ar^n = \frac{a(r^{n+1} - 1)}{r - 1} \quad \text{for } n \geq 1$$

- (b) Use the substitution $u = x \ln x$ to evaluate $\int_1^2 \frac{\ln x^2 + 2}{(x \ln x - 1)^2} dx$. 3

- (c) A population of 1200 feral cats is released into Darling Point. The rate of increase of the feral cat population is: $\frac{dP}{dt} = \frac{P}{30} \left(1 - \frac{P}{10000} \right)$ where P is the feral cat population and t is the number of months.

- (i) Show that $\frac{10000}{P(10000 - P)} = \frac{1}{P} + \frac{1}{10000 - P}$ 1

- (ii) Hence solve the differential equation $\frac{dP}{dt} = \frac{P}{30} \left(1 - \frac{P}{10000} \right)$ for P in terms of t . 3

- (iii) Hence find the limiting feral cat population. 1

- (iv) How many months does it take for the population to double? 1

- (d) Without the use of calculus, find the maximum and minimum points on the curve of $y = \frac{5}{2 + \sin x + \cos x}$ for $0 \leq x \leq 2\pi$ 3

End of Question 13

Question 14 (15 marks) Use the Question 14 Writing Booklet.

(a) Solve $\sin 7x + \sin x = \sin 4x$ for $0 \leq x \leq \frac{\pi}{2}$. **3**

(b) Find $\int_0^{\frac{\pi}{8}} \cos^4 x \, dx$. **3**

(c) Given that $f(x) = \cos^{-1}(2x-1) - 2\cos^{-1}\sqrt{x}$ for $0 \leq x \leq 1$, show that $f'(x) = 0$. **3**

(d) The ferry at Double Bay is being pulled into the wharf by a rope at a speed of 25m/min. The rope is attached to a point on the ferry 6m vertically below the level of the wharf. At what rate is the rope being drawn in, when the ferry is 28m from the wharf? **3**

(e) (i) Sketch the graph of $y = x^2 - 2x - 4$ **1**

(ii) Hence sketch the graph of $y = \frac{1}{|x^2 - 2x - 4|}$ **2**

End of Question 14

END OF EXAM

Ascham Maths Ext 1 Trial 2020

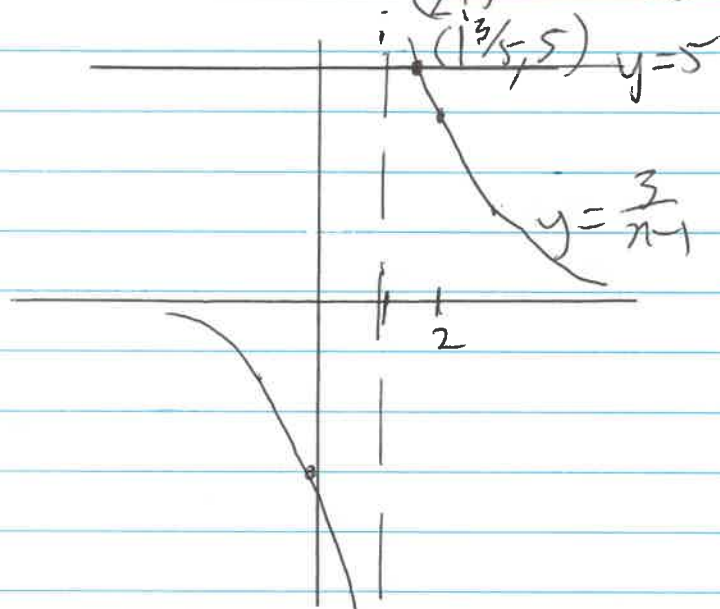
SOLUTIONS AND MARKING ALLOCATION

Section 1 Multiple Choice

1. A

2. B $P(\text{ASCHAM}) = \frac{1}{\binom{6!}{2!}} = \frac{1}{360}$

3.



$$\begin{aligned} \text{let } \frac{3}{x-1} &= 5 \\ \frac{3}{5} &= x-1 \\ x &= 1\frac{3}{5} \end{aligned}$$

D

4. $t = \tan \frac{\theta}{2} \quad \cos \theta = \frac{1-t^2}{1+t^2}, \quad \sin \theta = \frac{2t}{1+t^2}$

$$2\cos\theta + \sin\theta = -1$$

$$\frac{2(1-t^2)}{1+t^2} + \frac{2t}{1+t^2} = -1$$

D $2(1-t^2) + 2t = -(1+t^2)$

$$2 - 2t^2 + 2t = -1 - t^2$$

$$0 = t^2 - 2t - 3$$

$$0 = (t+1)(t-3)$$

$$\therefore t = -1 \quad \text{or} \quad t = 3$$

$$\begin{aligned}
 5 \quad \int \frac{x}{\sqrt{1-x^2}} dx &= -\frac{1}{2} \int -2x (1-x^2)^{-\frac{1}{2}} dx \\
 &= -\frac{1}{2} \int f'(x) (f(x))^{-\frac{1}{2}} dx \\
 &= -\frac{1}{2} \cdot 2 \left(f(x) \right)^{\frac{1}{2}} + C \\
 &= -\sqrt{1-x^2} + C
 \end{aligned}$$

6 C

7 original height when $y=4$

$$\begin{aligned}
 -3x^2 + 2x + 4 &= 4 \\
 -3x^2 + 2x &= 0 \\
 -x(3x-2) &= 0 \\
 x=0 &\quad \text{or} \quad x = \frac{2}{3}
 \end{aligned}$$

B.

$$\begin{aligned}
 8. \quad (3x-4)^6 &= {}^6C_0 (3x)^6 (-4)^0 + \dots + {}^6C_3 (3x)^3 (-4)^3 \\
 \text{Coefficient} &= {}^6C_3 \times 3^3 \times (-4)^3 \\
 &= -34560
 \end{aligned}$$

B.

9. B

$$10. \quad 8! \times 2! = 80640$$

C

Section 11

$$11 \text{ (a) } \underline{a} + \underline{b} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} + \begin{pmatrix} 4 \\ 7 \end{pmatrix} = \begin{pmatrix} 7 \\ 9 \end{pmatrix} \checkmark$$

$$\underline{a} - \underline{b} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} - \begin{pmatrix} 4 \\ 7 \end{pmatrix} = \begin{pmatrix} -1 \\ -5 \end{pmatrix} \checkmark$$

$$(ii) (\underline{a} + \underline{b}) \cdot (\underline{a} - \underline{b}) = \begin{pmatrix} 7 \\ 9 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ -5 \end{pmatrix}$$

$$= -7 + -45$$

$$= -52 \checkmark$$

$$\neq 0$$

$\therefore \underline{a} + \underline{b}$ is not perpendicular to $\underline{a} - \underline{b}$ $\checkmark$

$$(b) y = \sin^{-1}(2x)$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-(2x)^2}} \times 2 = \frac{2}{\sqrt{1-4x^2}} \checkmark$$

$$\text{when } x_1 = \frac{1}{4}, y_1 = \sin^{-1}(2 \times \frac{1}{4}) = \sin^{-1}(\frac{1}{2})$$

$$= \frac{\pi}{6}$$

$$m = \frac{2}{\sqrt{1-4(\frac{1}{4})^2}} = \frac{2}{\sqrt{\frac{3}{4}}} = \frac{4}{\sqrt{3}} \checkmark$$

$$y - y_1 = m(x - x_1)$$

$$y - \frac{\pi}{6} = \frac{4}{\sqrt{3}}(x - \frac{1}{4}) \checkmark$$

$$y = \frac{4}{\sqrt{3}}x + \frac{\pi}{6} - \frac{1}{\sqrt{3}}$$

$$(b) \quad P(x) = x^3 - 4x^2 + tx + 10$$

$$\alpha + (-\alpha) + \beta = 4$$

$$\beta = 4 \checkmark$$

$$\alpha(-\alpha)\beta = -10$$

$$-\alpha^2 \times 4 = -10$$

$$\alpha^2 = \frac{5}{2}$$

$$\alpha = \sqrt{\frac{5}{2}} \checkmark \text{ and } -\alpha = -\sqrt{\frac{5}{2}}$$

$$\alpha\beta + \alpha(-\alpha) + (-\alpha)\beta = t$$

$$-\alpha^2 = t$$

$$\therefore t = -\frac{5}{2}$$

$$(c) (i) \quad \ddot{x} = 0$$

$$\dot{x} = \int 0 dt \\ = C_1$$

$$\text{when } t=0, \dot{x} = 360$$

$$\therefore \dot{x} = 360 \quad \checkmark$$

$$\ddot{y} = -10$$

$$\dot{y} = \int -10 dt \\ = -10t + C_2$$

$$\text{when } t=0, \dot{y} = 0$$

$$\therefore \dot{y} = -10t \quad \checkmark$$

$$ii) \text{ when } t=0, x=0$$

$$x = \int 360 dt \\ = 360t + C_3$$

$$\text{when } t=0, x=0 \therefore C_3=0$$

$$\text{and } x = 360t \quad (1) \quad \checkmark$$

$$y = \int (-10t) dt \\ = -5t^2 + C_4$$

$$\text{when } t=0, y = 1.7 \text{ m}$$

$$\therefore C_4 = 1.7 \\ \text{and } y = -5t^2 + 1.7 \quad (2) \quad \checkmark$$

$$iii) \text{ from (1) } t = \frac{x}{360} \quad (3) \quad \text{sub (3) into (2):}$$

$$y = -5 \left(\frac{x}{360} \right)^2 + 1.7$$

$$= -\frac{x^2}{25920} + 1.7 \quad \checkmark \quad (4)$$

$$iv) \text{ when } x = 100, \quad y = \frac{-100^2}{25920} + 1.7$$

$$= 1.31 \text{ m (nearest cm)}$$

(v) From (i) above, when $n=100$,

$$100 = 360t$$

$$\therefore t = \frac{100}{360} = \frac{5}{18} \text{ second } \checkmark$$

$$\text{when } t = \frac{5}{18}, \dot{x} = 360, \dot{y} = -10 \times \frac{5}{18} = -\frac{50}{18}$$

$$\text{speed} = \sqrt{\dot{x}^2 + \dot{y}^2}$$

$$= \sqrt{360^2 + \left(-\frac{50}{18}\right)^2}$$

$$= 360.01 \text{ m/s (5 sig figs) } \checkmark$$

$$12(a) \frac{dy}{dx} = \frac{2x-1}{y^2}$$

$$\int y^2 dy = \int (2x-1) dx$$

$$\frac{y^3}{3} = x^2 - x + C \checkmark$$

$$\text{when } x=0, y = \sqrt[3]{3}$$

$$\text{so } \frac{(\sqrt[3]{3})^3}{3} = C \quad \text{or } C=1 \checkmark$$

$$\frac{y^3}{3} = x^2 - x + 1 \checkmark$$

$$y^3 = 3x^2 - 3x + 3$$

$$y = \sqrt[3]{3x^2 - 3x + 3}$$

$$(b) \quad R \cos(x+\alpha) = R \cos x \cos \alpha - R \sin x \sin \alpha$$

$$\text{let } R \cos(3x+\alpha) = R \cos x \cos 3x - R \sin x \sin 3x \\ = 3 \cos 3x - 4 \sin 3x$$

$$R^2 = 3^2 + 4^2 \\ = 5^2$$

$$\therefore R = 5 \quad \checkmark$$

$$5 \cos \alpha = 3$$

$$5 \sin \alpha = 4$$

$$\therefore \alpha = \cos^{-1}\left(\frac{3}{5}\right) \quad \checkmark$$

$$\therefore 5 \cos\left(3x + \cos^{-1}\frac{3}{5}\right) = 5$$

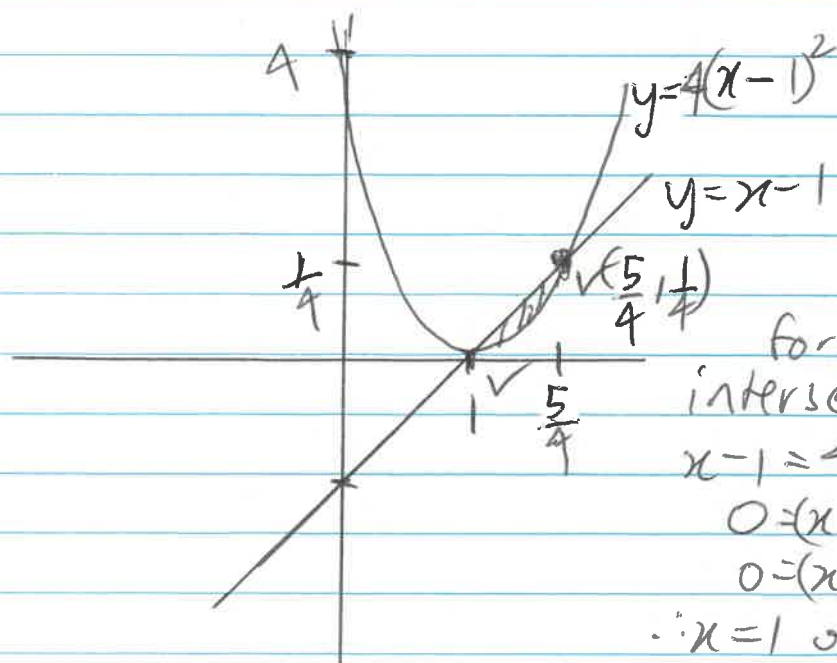
$$\cos\left(3x + \cos^{-1}\frac{3}{5}\right) = 1$$

$$3x + \cos^{-1}\frac{3}{5} = 0, 2\pi$$

$$3x = -\cos^{-1}\frac{3}{5}, 2\pi - \cos^{-1}\frac{3}{5}$$

$$x = \frac{2\pi}{3} - \frac{1}{3} \cos^{-1}\frac{3}{5} \quad \checkmark$$

(c) (i)



for point of intersection;

$$x-1 = 4(x-1)^2$$

$$0 = (x-1)(4(x-1)-1)$$

$$0 = (x-1)(4x-5)$$

$$\therefore x = 1 \text{ or } x = \frac{5}{4}$$

$$\text{when } x = 1, y = 1-1 = 0$$

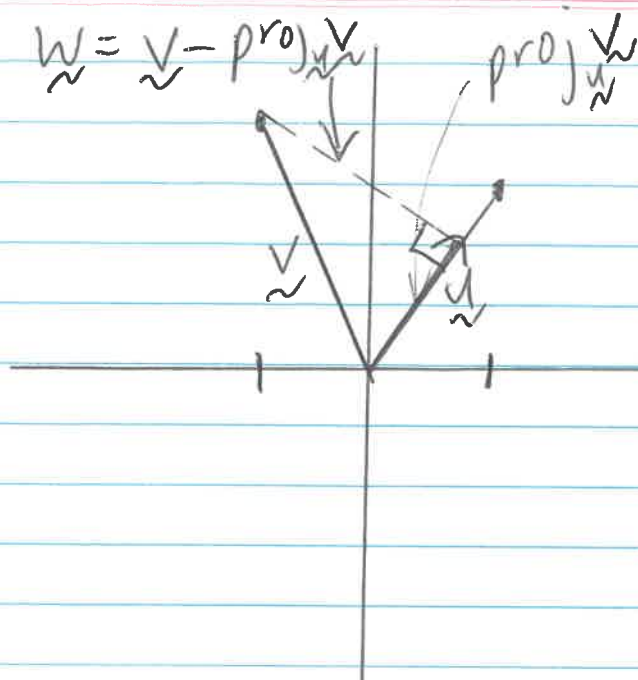
$$x = \frac{5}{4}, y = \frac{5}{4} - 1 = \frac{1}{4}$$

$$\begin{aligned}
 \text{(ii)} \quad V &= \pi \int_1^{\frac{5}{4}} (x-1)^2 dx - \pi \int_1^{\frac{5}{4}} (4(x-1)^2)^2 dx \checkmark \\
 &= \pi \left[\frac{(x-1)^3}{3} \right]_1^{\frac{5}{4}} - 16\pi \left[\frac{(x-1)^5}{5} \right]_1^{\frac{5}{4}} \checkmark \\
 &= \pi \left[\frac{(\frac{1}{4})^3}{3} \right] - 16\pi \left[\frac{(\frac{1}{4})^5}{5} \right] \\
 &= \frac{\pi}{480} \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \text{(d) i)} \quad \text{proj}_{\underline{u}} \underline{v} &= \frac{\underline{v} \cdot \underline{u}}{\underline{u} \cdot \underline{u}} \times \underline{u} & \underline{u} &= 2\hat{i} + 3\hat{j} \\
 & & \underline{v} &= -2\hat{i} + 4\hat{j} \\
 &= \frac{(-2 \times 2 + 3 \times 4)}{(2^2 + 3^2)} (2\hat{i} + 3\hat{j}) \\
 &= \frac{8}{13} (2\hat{i} + 3\hat{j}) \\
 &= \frac{16}{13} \hat{i} + \frac{24}{13} \hat{j} \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \text{ii)} \quad \underline{w} &= \underline{v} - \text{proj}_{\underline{u}} \underline{v} = -2\hat{i} + 4\hat{j} - \left(\frac{16}{13}\hat{i} + \frac{24}{13}\hat{j} \right) \\
 &= -\frac{42}{13}\hat{i} + \frac{28}{13}\hat{j} \checkmark
 \end{aligned}$$

(iii)



$\therefore \angle$ between w and $u = 90^\circ \checkmark$

13 (a) $a + ar + ar^2 + ar^3 + \dots + ar^n = \frac{a(r^{n+1} - 1)}{r - 1}$ ①

Step 1 Prove ① holds for $n=1$

$$\text{LHS} = a + ar$$

$$\text{RHS} = \frac{a(r^2 - 1)}{r - 1} = \frac{a(r-1)(r+1)}{(r-1)}$$

$$= a(r+1)$$

$$= a + ar = \text{LHS}$$

$\therefore$ ① holds for $n=1$

Step 2 Assume ① holds for $n=k$ and prove ① holds for $n=k+1$.

i.e.

$$a + ar + ar^2 + \dots + ar^k = \frac{a(r^{k+1} - 1)}{r - 1} \quad \text{②}$$

$$\text{RTP } a + ar + ar^2 + \dots + ar^k + ar^{k+1} = \frac{a(r^{k+1+1} - 1)}{r-1}.$$

$$\begin{aligned} \text{LHS} &= a + ar + ar^2 + \dots + ar^k + ar^{k+1} \\ &= \frac{a(r^{k+1} - 1)}{r-1} + ar^{k+1} \quad (\text{from (2)}) \checkmark \\ &= \frac{a(r^{k+1} - 1)}{r-1} + \frac{ar^{k+1}(r-1)}{r-1} \\ &= \frac{ar^{k+1} - a + ar^{k+2} - ar^{k+1}}{r-1} \\ &= \frac{ar^{k+2} - a}{r-1} \checkmark \\ &= \frac{a(r^{k+1+1} - 1)}{r-1} \\ &= \text{RHS as required.} \end{aligned}$$

$\therefore$ (i) holds for $n=k+1$ given it holds for $n=k$.

Step 3 It follows by mathematical induction from Step 1 and 2 that (i) holds for $n \geq 1$.

$$\begin{aligned} \text{(b)} \quad u &= x \ln x \\ \frac{du}{dx} &= x \left(\frac{d}{dx} (\ln x) \right) + \ln x \left(\frac{d}{dx} (x) \right) \\ &= x \times \frac{1}{x} + (\ln x) \times 1 \\ &= \ln x + 1 \\ \therefore du &= (\ln x + 1) dx \quad \checkmark \end{aligned}$$

When $x=1$, $u=1 \ln 1 = 0$

$$x=2e, u=2 \times \ln 2 = 2 (\ln 2) = 2 \ln 2$$

$$\int_1^2 \frac{\ln x^2 + 2}{(x \ln x - 1)^2} dx = \int_1^2 \frac{2 \ln x + 2}{(x \ln x - 1)^2} dx$$

$$= 2 \int_1^2 \frac{(\ln x + 1) dx}{(x \ln x - 1)^2}$$

$$= 2 \int_0^{2 \ln 2} \frac{du}{(u-1)^2}$$

$$= 2 \int_0^{2 \ln 2} (u-1)^{-2} du$$

$$= 2 \left[\frac{(u-1)^{-1}}{-1} \right]_0^{2 \ln 2}$$

$$= -2 \left[\frac{1}{2 \ln 2 - 1} - \frac{1}{0-1} \right]$$

$$= -2 \left[1 + \frac{1}{2 \ln 2 - 1} \right] \checkmark$$

$$= -2 \left[\frac{2 \ln 2 - 1}{2 \ln 2 - 1} + \frac{1}{2 \ln 2 - 1} \right]$$

$$= \frac{-2 (2 \ln 2)}{2 \ln 2 - 1}$$

$$= \frac{4 \ln 2}{1 - 2 \ln 2}$$

$$(c) (i) \frac{10000}{P(10000-P)} = \frac{1}{P} + \frac{1}{10000-P}$$

$$\begin{aligned} \text{RHS} &= \frac{1}{P} + \frac{1}{10000-P} = \frac{(10000-P) + P}{P(10000-P)} \checkmark \\ &= \frac{10000}{P(10000-P)} \\ &= \text{LHS} \end{aligned}$$

$$(ii) \frac{dP}{dt} = \frac{P}{30} \left(1 - \frac{P}{10000} \right) = \frac{P}{30} \left(\frac{10000-P}{10000} \right)$$

$$\int \frac{10000}{P(10000-P)} dP = \int \frac{1}{30} dt \checkmark$$

$$\int \left(\frac{1}{P} + \frac{1}{10000-P} \right) dP = \frac{1}{30} t + C$$

$$\ln P - \ln(10000-P) = \frac{1}{30} t + C \checkmark$$

$$\text{when } t=0, P=1200$$

$$\ln 1200 - \ln(10000-1200) = C$$

$$\therefore C = \ln \left(\frac{1200}{8800} \right)$$

$$= \ln \frac{3}{22}$$

$$\ln P - \ln(10000-P) = \frac{1}{30} t + \ln \frac{3}{22}$$

$$\ln \left(\frac{P}{10000-P} \right) = \frac{1}{30} t + \ln \frac{3}{22} \checkmark$$

$$\frac{P}{10000 - P} = e^{\left(\frac{1}{30}t + \ln\left(\frac{3}{22}\right)\right)}$$

$$\begin{aligned} P &= \left(e^{\left(\frac{1}{30}t\right)} \cdot e^{\ln\left(\frac{3}{22}\right)}\right)(10000 - P) \\ &= \frac{3}{22} e^{\frac{1}{30}t} (10000) - \frac{3}{22} e^{\frac{1}{30}t} (P) \end{aligned}$$

$$P + \frac{3}{22} e^{\frac{1}{30}t} P = \frac{3}{22} e^{\frac{1}{30}t} (10000)$$

$$P\left(1 + \frac{3}{22} e^{\frac{1}{30}t}\right) = \frac{30000}{22} e^{\frac{1}{30}t}$$

$$P = \frac{30000}{22 \left(1 + \frac{3}{22} e^{\frac{1}{30}t}\right)} \div e^{\frac{1}{30}t} \div e^{\frac{1}{30}t}$$

$$P = \frac{30000}{22\left(e^{-\frac{1}{30}t} + \frac{3}{22}\right)} \checkmark$$

iii) as $t \rightarrow \infty$, $P \rightarrow \frac{30000}{22 \times \frac{3}{22}} = 10000 \checkmark$

iv) when $t = 0$, $P = 1200$

population doubles when $P = 2400$

$$2400 = \frac{30000}{22\left(e^{-\frac{1}{30}t} + \frac{3}{22}\right)}$$

$$e^{-\frac{1}{30}t} + \frac{3}{22} = \frac{30000}{22 \times 2400} = \frac{25}{22 \times 2}$$

$$e^{-\frac{1}{30}t} = \frac{25}{22 \times 2} - \frac{6}{22 \times 2}$$

$$= \frac{19}{44}$$

$$e^{\frac{1}{30}t} = \frac{44}{19}$$

$$\frac{1}{30}t = \ln\left(\frac{44}{19}\right)$$

$$t = 30 \ln\left(\frac{44}{19}\right)$$

$$= 25.19 \dots$$

$$= 25 \text{ months (to the nearest month)}$$

(d) let $\cos x + \sin x = R \sin(x + \alpha)$
 $= R \sin \alpha \cos x + R \cos \alpha \sin x$

$$R \sin \alpha = 1$$

$$R \cos \alpha = 1$$

$$R = \sqrt{1^2 + 1^2} = \sqrt{2} \quad \text{and} \quad \cos \alpha = \sin \alpha = \frac{1}{\sqrt{2}}$$

$$\therefore \alpha = \frac{\pi}{4}$$

$$\therefore 2 + \sin x + \cos x$$

$$= 2 + \sqrt{2} \sin\left(x + \frac{\pi}{4}\right) \text{ has a max of } 2 + \sqrt{2}$$

$$\text{(when } x = \frac{\pi}{4}\text{)}$$

$$\text{and a min of } 2 - \sqrt{2}$$

$$\text{(when } x = \frac{5\pi}{4}\text{)}$$

$$\therefore \frac{5}{2 + \sin x + \cos x}$$

$$\text{has a max at } \frac{5}{2 - \sqrt{2}} \text{ when } x = \frac{5\pi}{4} \checkmark$$

and a min at $\frac{5}{2+\sqrt{2}}$ when $x = \frac{\pi}{4}$ ✓

14 (a) $\sin 7x + \sin x = \sin 4x$

now $\frac{1}{2}(\sin(4x+3x) + \sin(4x-3x)) = \sin 4x \cos 3x$ ✓

$\therefore \sin 7x + \sin x = 2 \sin 4x \cos 3x$

and $\sin 4x = 2 \sin 4x \cos 3x$

$0 = 2 \sin 4x \cos 3x - \sin 4x$

$0 = \sin 4x (2 \cos 3x - 1)$

$\therefore \sin 4x = 0$

$4x = 0, \pi, 2\pi, 3\pi, 4\pi$

$\therefore x = 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi$ ✓

or $2 \cos 3x - 1 = 0$

$\cos 3x = \frac{1}{2}$

$3x = \frac{\pi}{3}, 2\pi - \frac{\pi}{3}$

$3x = \frac{\pi}{3}, \frac{5\pi}{3}$

$x = \frac{\pi}{9}, \frac{5\pi}{9}$ ✓

(b) Recall $\cos 2x = 2 \cos^2 x - 1$

So $2 \cos^2 x = \cos 2x + 1$

$\cos^2 x = \frac{1}{2}(\cos 2x + 1)$

$\cos^4 x = \left(\frac{1}{2}(\cos 2x + 1)\right)^2$ ✓

$= \frac{1}{4}(\cos^2 2x + 2 \cos 2x + 1)$

$= \frac{1}{4}\left(\frac{1}{2}(\cos 4x + 1) + 2 \cos 2x + 1\right)$

$$\begin{aligned}
 \int_0^{\frac{\pi}{8}} \cos^4 x \, dx &= \frac{1}{4} \int_0^{\frac{\pi}{8}} \left(\frac{1}{2} \cos 4x + \frac{1}{2} + 2 \cos 2x + 1 \right) dx \\
 &= \frac{1}{8} \int_0^{\frac{\pi}{8}} (\cos 4x + 4 \cos 2x + 3) dx \\
 &= \frac{1}{8} \left[\frac{1}{4} \sin 4x + 2 \sin 2x + 3x \right]_0^{\frac{\pi}{8}} \checkmark \\
 &= \frac{1}{8} \left[\left(\frac{1}{4} \sin \frac{\pi}{2} + 2 \sin \frac{\pi}{4} + 3 \frac{\pi}{8} \right) - 0 \right] \\
 &= \frac{1}{8} \left(\frac{1}{4} + 2 \times \frac{1}{\sqrt{2}} + \frac{3\pi}{8} \right) \checkmark \\
 &= \frac{1}{32} \left(1 + 4\sqrt{2} + \frac{3\pi}{2} \right)
 \end{aligned}$$

(c) $f(x) = \cos^{-1}(2x-1) - 2\cos^{-1}(\sqrt{x})$

$$f'(x) = \frac{-1}{\sqrt{1-(2x-1)^2}} \times \frac{2}{1} \checkmark - 2 \frac{-1}{\sqrt{1-(\sqrt{x})^2}} \times \frac{1}{2} x^{-\frac{1}{2}} \checkmark$$

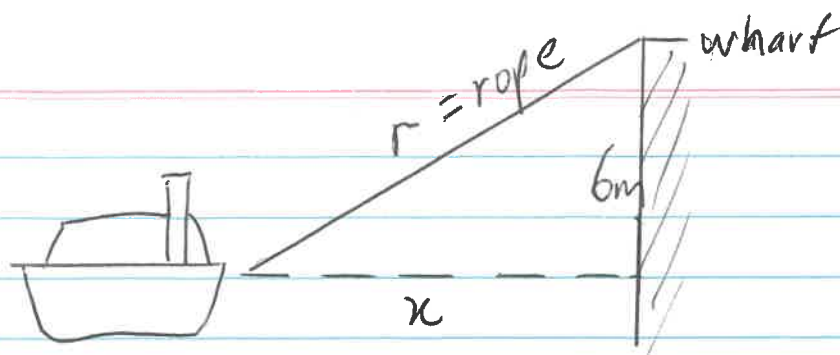
$$= \frac{-2}{\sqrt{1-(4x^2-4x+1)}} + \frac{1}{\sqrt{x}\sqrt{1-x}}$$

$$= \frac{-2}{\sqrt{4x-4x^2}} + \frac{1}{\sqrt{x}\sqrt{1-x}}$$

$$= \frac{-2}{\sqrt{4x(1-x)}} + \frac{1}{\sqrt{x}\sqrt{1-x}} \checkmark$$

$$= \frac{-2}{2\sqrt{x}\sqrt{1-x}} + \frac{1}{\sqrt{x}\sqrt{1-x}} = 0 \text{ as required.}$$

(d)



$$\frac{dx}{dt} = 25 \text{ m/min}$$

$$r^2 = x^2 + 6^2$$

$$r = \sqrt{x^2 + 36}$$

$$\therefore \frac{dr}{dx} = \frac{1}{2} \times 2x (x^2 + 36)^{-\frac{1}{2}}$$
$$= \frac{x}{\sqrt{x^2 + 36}} \quad \checkmark$$

$$\frac{dr}{dt} = \frac{dr}{dx} \times \frac{dx}{dt}$$

$$= \frac{x}{\sqrt{x^2 + 36}} \times 25 \quad \checkmark$$

$$\text{when } x = 28, \quad \frac{dr}{dt} = \frac{28}{\sqrt{28^2 + 36}} \times 25$$

$$= \frac{28 \times 25}{\sqrt{820}}$$

$$= \frac{700}{2\sqrt{205}} \quad \checkmark$$

$$= \frac{350\sqrt{205}}{205}$$

$$= \frac{70\sqrt{205}}{41} \text{ m/min}$$

(e)

$$\begin{aligned}
 y &= x^2 - 2x - 4 \\
 &= x^2 - 2x + 1 - 5 \\
 &= (x-1)^2 - 5
 \end{aligned}$$

when $y=0$,

$$(x-1)^2 = 5$$

$$x-1 = \pm\sqrt{5}$$

$$x = 1 \pm \sqrt{5}$$

